

Mathematics Stage 3 -- Year B -- Unit 38

Chance & Data

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly.
MA3-CHAN-01	Conducts chance experiments and quantifies the probability.
MA3-DATA-01	Constructs graphs using many-to-one scales (collects and organises data, and creates and interprets tables and graphs).
MA3-DATA-02	Interprets data displays, including timelines and line graphs.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
Data A + foundational chance (Chance B) Collect categorical and numerical data by observation or survey (Data A). Tabulate and organise collected data, with or without spreadsheets (Data A). Assign probabilities to outcomes (incl. not equally likely) using fractions and benchmark FDP (Chance B). Use the term frequency; distinguish the frequency of an outcome from its probability (Chance B).	Chance B & Data B (applying) Create random generators to follow specified probabilities or proportions. Compare observed and expected frequencies; explain differences and randomness over small/large trials. Discuss the fairness of simple chance games and the idea of randomness. Estimate the make-up of a large collection by sampling with replacement. Interpret side-by-side column graphs; compare displays by range, mode and shape of distribution. Interpret media/factual data displays; identify bias and misleading representations.	Extended -- Stage 4 Probability & Data List the sample space and find $P(\text{event}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ (MA4-PRO-C-01). Find observed probability as a relative frequency; use complementary events $P(\text{not } A) = 1 - P(A)$ (MA4-PRO-C-01). Classify variables (numerical/categorical, discrete/continuous) and choose a suitable graph (MA4-DAT-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students describe probabilities using fractions, use the terms frequency and randomness, and explain what data displays show.
Understanding and fluency	Students collect and tabulate data, assign probabilities as fractions, and read frequency tables and graphs.
Reasoning	Students reason about fairness and randomness, why observed results differ from expected, and where data may be biased or misleading.
Problem solving	Students design random generators, run chance experiments, sample with replacement, and (Stage 4) find theoretical and complementary probabilities.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key finding s	

Post-test / unit evaluation

Date	
Key finding s	
Studen ts who need further consol idation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 38 | NESA Mathematics K-10 Syllabus (2022)

Unit 38 -- Outcome Teaching Record Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Collect categorical and numerical data by observation or survey.	DATA-A: collect ordinal/nominal categorical and discrete numerical data through observation or survey MA3-DATA-01 Year A	Mini Lesson 1		
R2	Record and organise collected data in a table.	DATA-A: tabulate collected data with and without digital technologies such as spreadsheets MA3-DATA-01 Year A	Mini Lesson 2		
R3	Assign probabilities to outcomes using fractions and benchmark FDP.	CHAN-B: record outcomes (incl. not equally likely) and assign probabilities using fractions; use benchmark FDP for likelihood MA3-CHAN-01 Year B	Mini Lesson 3		
R4	Use the term frequency; distinguish frequency from probability.	CHAN-B: use the term frequency; distinguish the frequency of an outcome from the probability of an outcome MA3-CHAN-01 Year B	Mini Lesson 4		
Y1	Create random generators to follow set probabilities or proportions.	CHAN-B: create random generators to follow specified probabilities or proportions MA3-CHAN-01 Year B	Mini Lesson 5		
Y2	Compare observed and expected results; randomness over small/large trials.	CHAN-B: explain why observed frequencies differ from expected; discuss differences after small and large numbers of trials MA3-CHAN-01 Year B	Mini Lesson 6		
Y3	Discuss the fairness of simple chance games and randomness.	CHAN-B: discuss the fairness of simple games involving chance and the idea of randomness MA3-CHAN-01 Year B	Mini Lesson 7		
Y4	Estimate the make-up of a large collection by sampling with replacement.	CHAN-B: determine the likely make-up of a large collection by sampling objects and returning them before the next sample MA3-CHAN-01 Year B	Mini Lesson 8		
Y5	Interpret and compare data displays (column graphs, range, mode, shape).	DATA-B: interpret side-by-side column graphs; compare displays by range, mode and shape of distribution MA3-DATA-02 Year B	Mini Lesson 9		
Y6	Interpret media data; identify bias and misleading displays.	DATA-B: interpret data in digital media/factual texts; identify sources of bias and misleading representations MA3-DATA-02 Year B	Mini Lesson 10		
G1	List the sample space and find $P(\text{event}) = \text{favourable} / \text{total}$.	Stage 4 PRO: list the sample space; $P(\text{event}) = \text{favourable outcomes} / \text{total outcomes}$; probabilities range 0-1 MA4-PRO-C-01 Stage 4 (Core)	Mini Lesson 11		
G2	Relative frequency and complementary events $P(\text{not } A) = 1 - P(A)$.	Stage 4 PRO: observed probability as relative frequency; complementary events sum to 1; solve complementary-event problems MA4-PRO-C-01 Stage 4 (Core)	Mini Lesson 12		

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
G3	Classify variables and choose a suitable graph.	Stage 4 DAT: classify variables numerical/categorical, discrete/continuous; select the best graph and justify MA4-DAT-C-01 Stage 4 (Core)	Mini Lesson 13		

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Unit 38 -- Mini Lesson Sequence

Chance & Data | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-4

Yellow band -- On-level -- Mini Lessons 5-10

Green band -- Extension -- Mini Lessons 11-13

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 38. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Collecting data by observation and survey **[HANDS-ON]**

R1 | Red band | DoE Lesson 2 (adapted)

Duration: 25-30 min | **Outcomes:** R1 | Syllabus: MA3-DATA-01, MAO-WM-01 | DATA-A: collect categorical and numerical data

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Data collection and probability | Key resources: Survey question cards, tally sheets, clipboards

Physical materials needed: tally sheets, survey question cards

Learning intention

Students are learning to:
collect categorical and numerical data by observation or by conducting a survey

Success criteria

Students can:

- * I can write a survey question that gives data that varies.
- * I can collect data by observing or asking a survey.
- * I can tell the difference between categorical and numerical data.

Mini masterclass structure (25-30 min)

Activate (4 min): Ask: 'What is your favourite sport?' vs 'How tall am I?' Which gives data that varies (a statistical question)?

Model (10 min): Demonstrate a quick class survey with a tally; show categorical data (favourite colour) and numerical data (number of pets).

Guided practice (10 min): Students collect data for a chosen question by observation or survey and tally the results.

Check (4 min): Exit question: is 'favourite fruit' categorical or numerical data?

Key vocabulary

data, categorical, numerical, observation, survey, tally, statistical question

Differentiation

Support: Provide ready-made survey questions and a pre-drawn tally sheet.
Extension: Compare data collected by observation with data collected by survey and discuss reliability.

Assessment

Teaching notes (annotate here)

Students collect data — links to R1 in the SOLO Show task.

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Organising data in a table

R2 | Red band | DoE Lesson 5 (adapted)

Duration: 25 min | **Outcomes:** R2 | Syllabus: MA3-DATA-01, MAO-WM-01 | DATA-A: tabulate collected data

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Conducting chance experiments (recording data) | Key resources: Frequency-table templates, spreadsheet (optional)

Learning intention		Success criteria	
Students are learning to: record and organise collected data in a frequency table		Students can: * I can put collected data into a tally and frequency table. * I can read how many times each result occurred from the table. * I can organise data with or without a spreadsheet.	
Mini masterclass structure (25 min)			
Activate (3 min): Show a messy list of results. Ask: how could we organise this so it's easy to read?			
Model (10 min): Turn a tally into a frequency table; add a total row. Show the same data in a spreadsheet column.			
Guided practice (8 min): Students organise a set of results into a frequency table and check the total.			
Check (4 min): Exit question: from a frequency table, which result happened most often?			
Key vocabulary		Differentiation	
table, tally, frequency, frequency table, total, spreadsheet, organise		Support: Provide a part-completed frequency table to finish. Extension: Build the frequency table in a spreadsheet and create a quick column graph from it.	
Assessment		Teaching notes (annotate here)	
Students tabulate data — links to R2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 3 Probability as a fraction **[HANDS-ON]**

R3 | Red band | DoE Lesson 1 (adapted)

Duration: 25-30 min

Outcomes: R3

Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: assign probabilities using fractions; benchmark FDP

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Fraction chance generators | Key resources: Spinners, bags of coloured counters, dice

Physical materials needed: spinners, bags of counters, dice

Learning intention

Students are learning to:
assign a probability to an outcome using a fraction, and say how likely it is

Success criteria

Students can:

- * I can write the probability of an outcome as a fraction (favourable ÷ total).
- * I can find probabilities when outcomes are not equally likely.
- * I can use benchmarks ($\frac{1}{2}$, $\frac{1}{4}$, 50%, 0.5) to describe how likely an outcome is.

Mini masterclass structure (25-30 min)

Activate (4 min): Show a bag with 3 red and 1 blue counter. Ask: what is the chance of red? Of blue?

Model (10 min): $P(\text{red}) = \frac{3}{4}$, $P(\text{blue}) = \frac{1}{4}$. Connect to benchmarks: $\frac{3}{4} = 75\%$; $\frac{1}{2} = 50\% = 0.5$. Show a spinner with unequal sections.

Guided practice (10 min): Students assign probabilities as fractions for spinners and bags where outcomes are not equally likely.

Check (4 min): Exit question: a bag has 2 green and 3 yellow. What is $P(\text{green})$?

Key vocabulary

probability, outcome, fraction, favourable, likely, benchmark, per cent

Differentiation

Support: Use a small number of equally likely outcomes first (e.g. 4 sections).
Extension: Find $P(\text{not an outcome})$ and show the probabilities of all outcomes add to 1.

Assessment

Students assign probabilities as fractions — links to R3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Frequency and probability

R4 | Red band | DoE Lessons 3 & 4 (adapted)

Duration: 25 min | **Outcomes:** R4 | Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: the term frequency; frequency vs probability

Linked to: DoE Lessons 3 & 4 (adapted) -- Lessons 3 & 4 -- Compare observed frequencies with expected results | Key resources: Dice, tally sheets

Learning intention		Success criteria	
Students are learning to: use the term frequency and tell the difference between frequency and probability		Students can: * I can use the word frequency for how many times an outcome happens. * I know probability is how likely an outcome is, written as a fraction. * I can tell the difference between the frequency and the probability of an outcome.	
Mini masterclass structure (25 min)			
Activate (3 min): Roll a die 10 times and tally. Ask: how many times did a 6 come up (frequency)? Model (10 min): Frequency = the count from the experiment. Probability = the expected chance, $P(6) = 1/6$. Compare the two for the same outcome. Guided practice (8 min): Students record frequencies from a short experiment and state the probability of each outcome. Check (4 min): Exit question: if a 3 came up 4 times in 12 rolls, what is its frequency? Its probability?			
Key vocabulary		Differentiation	
frequency, probability, outcome, count, trial, experiment, likely		Support: Keep the experiment short (e.g. 6 trials) and tally together. Extension: Convert frequencies to a relative frequency fraction (count ÷ total trials).	
Assessment		Teaching notes (annotate here)	
Students distinguish frequency from probability — links to R4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 5 Creating random generators [HANDS-ON]

Y1 | Yellow band | DoE Lesson 2 (adapted)

Duration: 30 min | **Outcomes:** Y1 | Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: create random generators for set probabilities

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Data collection and probability | Key resources: Blank spinners, counters, bags

Physical materials needed: blank spinners, bags, counters

Learning intention	Success criteria
Students are learning to: design a random generator that follows a set of probabilities or proportions	Students can: * I can design a spinner or bag to match given probabilities. * I can choose how many of each colour to use for a set proportion. * I can check my generator gives the probabilities I wanted.

Mini masterclass structure (30 min)

Activate (4 min): Challenge: design a spinner where red is twice as likely as blue. How?

Model (12 min): For $P(\text{red}) = 2/3$ and $P(\text{blue}) = 1/3$, make a spinner with 2 red sections and 1 blue, or a bag with 4 red and 2 blue counters.

Guided practice (10 min): Students design generators (spinners/bags) to match given probabilities and explain their choices.

Check (4 min): Exit question: design a bag where $P(\text{green}) = 1/4$.

Key vocabulary	Differentiation
random generator, spinner, probability, proportion, sections, outcome, design	Support: Provide a spinner already divided into equal sections to colour. Extension: Design a generator with three outcomes in the ratio 1 : 2 : 3.
Assessment	Teaching notes (annotate here)
Students create random generators — links to Y1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Observed vs expected results [HANDS-ON]

Y2 | Yellow band | DoE Lessons 3 & 4 (adapted)

Duration: 30 min | **Outcomes:** Y2 | Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: observed vs expected; randomness; small/large trials

Linked to: DoE Lessons 3 & 4 (adapted) -- Lessons 3 & 4 -- Compare observed frequencies with expected results | Key resources: Dice, coins, tally sheets

Physical materials needed: dice, coins, tally sheets

Learning intention		Success criteria	
Students are learning to: compare observed results with expected results and explain why they differ		Students can: * I can predict the expected results of a chance experiment. * I can run the experiment and compare observed with expected. * I can explain that with more trials the observed results get closer to expected (randomness).	
Mini masterclass structure (30 min)			
Activate (4 min): Flip a coin 10 times. Did you get exactly 5 heads? Why might it differ from expected? Model (12 min): Expected: P(heads) = ½, so ~5 of 10. Observed may be 6 or 3 — that's randomness. Combine the class results (many trials) and see it get closer to half. Guided practice (10 min): Students run a dice/coin experiment with small then large numbers of trials and compare observed with expected. Check (4 min): Exit question: why do more trials usually make observed results closer to expected?			
Key vocabulary		Differentiation	
observed, expected, frequency, probability, trial, randomness, sample size		Support: Pool results with a partner to reach more trials quickly. Extension: Graph how the observed proportion of heads changes as trials increase.	
Assessment		Teaching notes (annotate here)	
Students compare observed and expected results — links to Y2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 7 Fairness and randomness in chance games **[HANDS-ON]**

Y3 | Yellow band | DoE Lesson 4 (adapted)

Duration: 25-30 min **Outcomes:** Y3 Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: fairness of chance games; randomness

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Fairness, chance and probability | Key resources: Dice, spinners, coloured counters

Physical materials needed: dice, spinners, counters

Learning intention

Students are learning to:
discuss whether a simple chance game is fair using the idea of randomness

Success criteria

Students can:

- * I can work out each player's probability of winning.
- * I can decide whether a game is fair (equal chances) or unfair.
- * I can suggest how to make an unfair game fair.

Mini masterclass structure (25-30 min)

Activate (4 min): Game: Player A wins on 1-2, Player B on 3-6. Is this fair? Why not?

Model (10 min): $P(A) = 2/6$, $P(B) = 4/6$ — unfair. Discuss how to make it fair (A wins on 1-3, B on 4-6). Link to randomness.

Guided practice (10 min): Students play simple chance games, work out the probabilities, and judge fairness.

Check (4 min): Exit question: is a game where you win only on a 6 fair for both players? Explain.

Key vocabulary

fair, unfair, chance, probability, randomness, equal, outcome

Differentiation

Support: Use games with two equally likely outcomes first (coin flip).
Extension: Design a fair three-player game and prove it is fair using probabilities.

Assessment

Students judge fairness of chance games — links to Y3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Sampling with replacement [HANDS-ON]

Y4 | Yellow band | DoE Lesson 5 (adapted)

Duration: 30 min | **Outcomes:** Y4 | Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: estimate a large collection by sampling with replacement

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Conducting chance experiments with small and large numbers of trials | Key resources: Bag of mixed coloured counters, tally sheets

Physical materials needed: bag of mixed counters, tally sheets

Learning intention	Success criteria
Students are learning to: estimate the make-up of a large hidden collection by sampling with replacement	Students can: * I can take a sample, record it, and return it before sampling again. * I can use the proportions in my samples to estimate the whole collection. * I know more samples give a better estimate.

Mini masterclass structure (30 min)

Activate (4 min): Show a sealed bag of counters. Ask: without tipping it out, how could we estimate how many are red?

Model (12 min): Draw one counter, record the colour, put it back, repeat. After 20 draws, if 12 were red, estimate $\sim 12/20 = 3/5$ of the bag is red.

Guided practice (10 min): Students sample with replacement, tally results, and estimate the make-up of the collection.

Check (4 min): Exit question: why do we put the counter back each time before sampling again?

Key vocabulary	Differentiation
sample, sampling with replacement, proportion, estimate, collection, frequency	Support: Use a bag with only two colours and a small number of draws. Extension: Compare estimates from 10 samples vs 50 samples and discuss which is more reliable.

Assessment	Teaching notes (annotate here)
Students sample with replacement — links to Y4 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Interpreting and comparing data displays

Y5 | Yellow band | DoE Lessons 7 & 8 (adapted)

Duration: 30 min **Outcomes:** Y5 Syllabus: MA3-DATA-02, MAO-WM-01 | DATA-B: side-by-side column graphs; range, mode, shape

Linked to: DoE Lessons 7 & 8 (adapted) -- Lessons 7 & 8 -- Interpreting and comparing a range of data displays | Key resources: Printed side-by-side column graphs, data sets

Learning intention		Success criteria	
Students are learning to: interpret side-by-side column graphs and compare displays using range, mode and shape		Students can: * I can read values from a side-by-side column graph for two groups. * I can find the range and the mode (most frequent value) of a data set. * I can compare two displays by their shape, range and mode.	
Mini masterclass structure (30 min)			
Activate (4 min): Show a side-by-side column graph (boys vs girls favourite sport). Ask a comparison question. Model (12 min): Read values for each group; find the mode (tallest column) and the range (highest – lowest). Compare the shape of two displays. Guided practice (10 min): Students interpret side-by-side column graphs and compare data sets by range and mode. Check (4 min): Exit question: for 3, 5, 5, 8, what is the mode and the range?			
Key vocabulary		Differentiation	
column graph, side-by-side, range, mode, distribution, shape, compare		Support: Provide the range and mode definitions and a worked example to follow. Extension: Compare three displays and describe how the shape of the distribution differs.	
Assessment		Teaching notes (annotate here)	
Students interpret and compare data displays — links to Y5 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 10 Bias and misleading data in the media

Y6 | Yellow band | DoE Lessons 6 & 7 (adapted)

Duration: 25-30 min **Outcomes:** Y6 Syllabus: MA3-DATA-02, MAO-WM-01 | DATA-B: interpret media data; bias; misleading displays

Linked to: DoE Lessons 6 & 7 (adapted) -- Lessons 6 & 7 -- Interpret data presented in digital media and elsewhere | Key resources: Media graphs (newspaper/online), factual texts

Learning intention

Students are learning to:
interpret data displays in the media and identify bias or misleading features

Success criteria

Students can:

- * I can read and explain a data display from the media or a factual text.
- * I can spot a source of bias in how data is shown.
- * I can recognise when a graph is misleading (e.g. a stretched or cut axis).

Mini masterclass structure (25-30 min)

Activate (4 min): Show a graph with a y-axis that doesn't start at 0. Ask: does this make a small difference look big?

Model (10 min): Read a media graph; point out misleading features (broken axis, uneven scale, cherry-picked data) and possible bias.

Guided practice (10 min): Students analyse media data displays and identify bias or misleading features.

Check (4 min): Exit question: name one way a graph can be made to mislead the reader.

Key vocabulary

media, data display, bias, misleading, axis, scale, interpret

Differentiation

Support: Provide clearly misleading examples with the feature highlighted.
Extension: Redraw a misleading graph fairly and explain what changed.

Assessment

Students identify bias and misleading data — links to Y6 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Sample space and theoretical probability

G1 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes: G1** | Syllabus: MA4-PRO-C-01, MAO-WM-01 | Stage 4: sample space; $P(\text{event}) = \text{favourable} / \text{total}$

Learning intention		Success criteria
Students are learning to: list the sample space and find $P(\text{event}) = \text{favourable outcomes} / \text{total outcomes}$		Students can: * I can list the sample space (all possible outcomes) for an experiment. * I can write $P(\text{event}) = \text{favourable outcomes} \div \text{total outcomes}$. * I know probabilities range from 0 (impossible) to 1 (certain).
Mini masterclass structure (30 min)		
Activate (4 min): Roll a die. What are all the possible outcomes? That set is the sample space. Model (12 min): Sample space {1,2,3,4,5,6}. $P(\text{even}) = 3/6 = \frac{1}{2}$. Show the 0-to-1 probability scale; impossible = 0, certain = 1. Guided practice (10 min): Students list sample spaces and calculate $P(\text{event})$ as a fraction for dice, spinners and cards. Check (4 min): Exit question: a spinner has 8 equal sections, 3 red. What is $P(\text{red})$?		
Key vocabulary		Differentiation
sample space, outcome, event, favourable, probability, impossible, certain		Support: Provide the sample space already listed so students focus on $P(\text{event})$. Extension: Verify that the probabilities of all outcomes in the sample space add to 1.
Assessment		Teaching notes (annotate here)
Students find theoretical probabilities — extends R3.		Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Relative frequency and complementary events

G2 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G2 | Syllabus: MA4-PRO-C-01, MAO-WM-01 | Stage 4: relative frequency; complementary events

Learning intention		Success criteria
Students are learning to: find an observed probability as a relative frequency and use complementary events		Students can: * I can find an observed (experimental) probability as a relative frequency (count ÷ trials). * I can find the complement of an event. * I can use $P(\text{not } A) = 1 - P(A)$.
Mini masterclass structure (30 min)		
Activate (4 min): In 50 spins, red came up 20 times. What is the observed probability of red? Model (12 min): Relative frequency = $20/50 = 2/5$. The complement of 'red' is 'not red'; $P(\text{not red}) = 1 - 2/5 = 3/5$. They add to 1. Guided practice (10 min): Students calculate relative frequencies and complementary probabilities from experiments. Check (4 min): Exit question: if $P(\text{win}) = 1/4$, what is $P(\text{not win})$?		
Key vocabulary		Differentiation
relative frequency, observed probability, complement, complementary events, trials		Support: Provide the trial counts so students focus on forming the fraction. Extension: Solve a multi-step problem combining relative frequency and complementary events.
Assessment		Teaching notes (annotate here)
Students use relative frequency and complements — extends Y2.		Taught as described / adapted / replaced -- note here:

Mini Lesson 13 Classify data and choose the right graph

G3 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G3 | Syllabus: MA4-DAT-C-01, MAO-WM-01 | Stage 4: classify variables; choose a suitable graph

Learning intention		Success criteria	
Students are learning to: classify variables as numerical or categorical and choose a graph that suits the data		Students can: * I can classify a variable as numerical (discrete/continuous) or categorical (nominal/ordinal). * I can choose a graph that suits the type of data. * I can justify why my chosen graph is appropriate.	
Mini masterclass structure (30 min)			
Activate (4 min): Sort: shoe size, eye colour, height, favourite subject — which are numerical, which categorical?			
Model (12 min): Numerical = numbers (discrete: counts; continuous: measures). Categorical = groups (nominal/ordinal). Match each to a sensible graph (column graph, dot plot, line graph).			
Guided practice (10 min): Students classify variables and choose (and justify) a suitable graph for each.			
Check (4 min): Exit question: 'number of pets' — numerical or categorical? Which graph suits it?			
Key vocabulary		Differentiation	
variable, numerical, categorical, discrete, continuous, nominal, ordinal, graph		Support: Provide a sorting table with the categories labelled. Extension: Justify why one graph type would mislead for a given data set.	
Assessment		Teaching notes (annotate here)	
Students classify data and choose graphs — extends R1/R2/Y5.		Taught as described / adapted / replaced -- note here:	

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Unit 38 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Collect categorical and numerical data by observation or survey.		
Video	What Is a Statistical Question? Math Defined	https://www.youtube.com/watch?v=QZjbwzbCCNE
Video	Surveys in Statistics LearnFree	https://www.youtube.com/watch?v=Y_EQfy-uk0Y
Worksheet (print)	Corbettmaths — Bar Charts / Tally Charts / Pictograms (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/bar-charts-tally-charts-and-pictograms-pdf.pdf
R2 -- Record and organise collected data in a table.		
Video	How to Make a Frequency Table Math with Mr. J	https://www.youtube.com/watch?v=dcUiY2CidIQ
Video	How to Read a Frequency Table Math with Mr. J	https://www.youtube.com/watch?v=g_pyxyJigM8
Worksheet (print)	Corbettmaths — Drawing Bar Charts (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/drawing-bar-charts-pdf.pdf
R3 -- Assign probabilities to outcomes using fractions and benchmark FDP.		
Video	An Intro to Probability Basic Probability Math with Mr. J	https://www.youtube.com/watch?v=6ldNYdA6no8
Video	Finding Probability Basic Probability Math with Mr. J	https://www.youtube.com/watch?v=QEdHtF1CKRQ
Worksheet (print)	Corbettmaths — Probability (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf
R4 -- Use the term frequency; distinguish frequency from probability.		
Video	Theoretical vs. Experimental Probability Math with Mr. J	https://www.youtube.com/watch?v=Ay5pnC-0bVI
Video	Frequency Tables and Dot Plots Math with Mr. J	https://www.youtube.com/watch?v=ZQ5cXMx3Db0
Worksheet (print)	Corbettmaths — Relative Frequency (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf
Y1 -- Create random generators to follow set probabilities or proportions.		
Video	An Intro to Finding Probability Math with Mr. J	https://www.youtube.com/watch?v=zE-Y_QCHot8
Video	Theoretical Probability — Possible Outcomes Using A Spinner mrmasonet	https://www.youtube.com/watch?v=3QmNvODInt8
Worksheet (print)	Corbettmaths — Probability (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf
Y2 -- Compare observed and expected results; randomness over small/large trials.		
Video	Theoretical vs. Experimental Probability Math with Mr. J	https://www.youtube.com/watch?v=Ay5pnC-0bVI
Video	Experimental vs. Theoretical Probability Explained! Mario's Math Tutoring	https://www.youtube.com/watch?v=J0ijhF-cdFU
Worksheet (print)	Corbettmaths — Relative Frequency (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf
Y3 -- Discuss the fairness of simple chance games and randomness.		
Video	Fair and Unfair Games Andrea Knechtel	https://www.youtube.com/watch?v=-X5vAmWQDuE
Video	Fair and Unfair Games (Project Maths) TheMathsTutor.ie	https://www.youtube.com/watch?v=pDOfqsdZ7Hs
Worksheet (print)	Corbettmaths — Probability (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf
Y4 -- Estimate the make-up of a large collection by sampling with replacement.		
Video	Sampling and Estimation Dietmar Schaffner	https://www.youtube.com/watch?v=XiU7srCwokU
Video	Probability — Drawing Counters from a Bag Tambuwal Maths Class	https://www.youtube.com/watch?v=-4JgsyckoLM
Worksheet (print)	Corbettmaths — Relative Frequency (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf
Y5 -- Interpret and compare data displays (column graphs, range, mode, shape).		

Type	Resource	Link
Video	Averages (Mean/Median/Mode) & Range Math with Mr. J	https://www.youtube.com/watch?v=e3uY2LraXts
Video	Reading Graphs (Simplifying Math) Buffington	https://www.youtube.com/watch?v=C0-eJfOg3nc
Worksheet (print)	Corbettmaths — Two-Way Tables (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/two-way-tables-pdf.pdf
Y6 -- Interpret media data; identify bias and misleading displays.		
Video	How to Spot a Misleading Graph TED-Ed (Lea Gaslowitz)	https://www.youtube.com/watch?v=E91bGT9BjYk
Video	Misleading Graphs Corbettmaths	https://www.youtube.com/watch?v=-rYUi_dfhY0
Worksheet (print)	Corbettmaths — Reading Bar Charts (PDF)	https://corbettmaths.com/wp-content/uploads/2018/12/Reading-Bar-Charts.pdf
G1 -- List the sample space and find P(event) = favourable / total.		
Video	Introduction to Probability — Sample Space & Tree Diagrams The Organic Chemistry Tutor	https://www.youtube.com/watch?v=SkidyDQuupA
Video	An Intro to Probability Basic Probability Math with Mr. J	https://www.youtube.com/watch?v=6IdNYdA6no8
Worksheet (print)	Corbettmaths — Listing Outcomes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/listing-outcomes-pdf.pdf
G2 -- Relative frequency and complementary events $P(\text{not } A) = 1 - P(A)$.		
Video	Complementary Events in Probability Math and Stats Help	https://www.youtube.com/watch?v=hJbj7UQIJNw
Video	Theoretical vs. Experimental Probability Math with Mr. J	https://www.youtube.com/watch?v=Ay5pnC-0bVI
Worksheet (print)	Corbettmaths — Relative Frequency (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf
G3 -- Classify variables and choose a suitable graph.		
Video	Types of Data: Categorical and Numerical EJFullerton	https://www.youtube.com/watch?v=Vem8Ro-cHIU
Video	Types of Data: Categorical vs Numerical Data 365 Data Science	https://www.youtube.com/watch?v=DUCXZ08IdMo
Worksheet (print)	Corbettmaths — Averages and Range (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/averages-and-range-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	Introduction to Probability — Sample Space & Tree Diagrams The Organic Chemistry Tutor	https://www.youtube.com/watch?v=SkidyDQuupA
Video	Complementary Events in Probability Math and Stats Help	https://www.youtube.com/watch?v=hJbj7UQIJNw
Video	Types of Data: Categorical vs Numerical Data 365 Data Science	https://www.youtube.com/watch?v=DUCXZ08IdMo
Worksheet (print)	Corbettmaths — Listing Outcomes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/listing-outcomes-pdf.pdf
Worksheet (print)	Corbettmaths — Relative Frequency (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Bar Charts / Tally Charts / Pictograms (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/bar-charts-tally-charts-and-pictograms-pdf.pdf>
- ☐ Corbettmaths — Drawing Bar Charts (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/drawing-bar-charts-pdf.pdf>
- ☐ Corbettmaths — Probability (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf>
- ☐ Corbettmaths — Relative Frequency (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf>
- ☐ Corbettmaths — Two-Way Tables (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/two-way-tables-pdf.pdf>
- ☐ Corbettmaths — Reading Bar Charts (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/12/Reading-Bar-Charts.pdf>
- ☐ Corbettmaths — Listing Outcomes (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/listing-outcomes-pdf.pdf>
- ☐ Corbettmaths — Averages and Range (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/averages-and-range-pdf.pdf>